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REVIEW

Analysis of soil and water pollution produced by improper medical waste disposal and its impacts

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Abstract Many countries struggle with the systematic separation and disposal of medical waste, which poses significant risks of disease transmission and cross-contamination. This study synthesizes literature from 2017 to 2023 to examine the environmental consequences of improper healthcare waste management, specifically regarding soil and water ecosystems. The review establishes a direct scientific link between medical waste dumping sites and the presence of heavy metal concentrations that exceed safety thresholds in surrounding agricultural lands. A key finding of this research is the identification of a "toxic pathway" where contaminated soil and water lead to significant nutritional depletion in crops and subsequent bioaccumulation of toxins in humans and animals. Furthermore, the analysis reveals that while technical solutions exist, the primary barrier to environmental safety is a systemic failure in policy enforcement and segregation protocols within healthcare sectors. These findings provide a categorized framework for future research to quantify the long-term impact of medical pollutants on food safety and provide a foundation for developing more stringent, evidence-based waste management policies to mitigate ecological degradation.

Keywords Medical disposal · Hazardous waste · Pollution · Diseases

Introduction

Expansion in economic and growth in population leads to an increase in solid waste production. The fast-growing industry in the current era is the healthcare sector, due to the demand in medical treatments which results in an increase in need for medical waste disposal and treatment (Priyadarshini & Kenny, 2021). In recent years, there is increase in waste generation as there is an advancement in the medical sector across the globe which has increased in the medical waste. The waste related to the healthcare sector contributes an expanded category of solid waste mainly produced in laboratories, hospitals, dental offices and liquid waste in blood banks (Torkashvand et al., 2022). The waste from hospitals is generally from clinics, pharmacies and other healthcare centres. The hospital waste generally contains chemical waste, infectious waste, sharp item, heavy metal and also materials that are radioactive or genotoxic. The waste in the hospital could affect the groundwater (GW), soil, air, surface water, surroundings and human health. There is advancement in technologies like incineration and autoclaving for disposing of hospital waste in developed countries (Mostafavi et al., 2022). Whereas in the countries that are developing, there is no

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adequate attention in the waste management in the hospital; also, the disposal of hospital waste is done together with the domestic waste (Shammi et al., 2022).

There is a worldwide concern regarding the medical waste management. The medical waste (MW) is generated from all the healthcare sectors that are managed poorly which in turn affects human, wildlife and the environment. A question arises among the researchers whether there is a reduction in waste disposal, as there is an increase in healthcare centres with an increase in population. There has been always a threat for the environment and human beings based on the waste generated by the activities of humans (Budjav, 2022). There is a need for a proper waste management system to avoid threat and for the betterment of the lifestyle of humans. The need for management of the medical waste has increased in the pandemic period as it might create significant danger for health in the human being and the environment (Barua & Hossain, 2021). In order to deal with waste management in the healthcare industry, it is significant to apply a circular economy and life cycle assessment to pave a way for green circulation of medical disposal (Antoniadou et al., 2021). There are many hazardous substances in the MW; poor management of MW leads to the risk on human health (Organization & Fund, 2025). There is often downplay in the MW management, yet the pandemic outbreak has made changes in the worldwide dynamics of MW (E. Singh et al., 2022a, 2022b). There is no proper management in MW; less than 60% of developed countries have no significant handling of waste. This potentially exposes healthcare workers to different pathogenic microorganisms while simultaneously affecting the environment in poorly managed waste disposal sites (WHO, 2022).

The threat of MW which is not properly managed leads to various transmission diseases; hence, 5.2 million people die every year (Chowdhury et al., 2022). Based on the existing literature, mishandling of MW by healthcare workers leads to diseases like tuberculosis, hepatitis C, AIDS and hepatitis B. This is mainly due to the mixing of MW with the domestic waste. Mainly developing countries have inadequate training on handling the MW (Khalid et al., 2021). In the pandemic period, nearly 49 percentage of people disposed of both the household waste and the COVID waste in the same trash container which increased the rate of infection among healthcare workers (Shammi et al., 2021). Correspondingly, the pathogenic organisms from this type of waste enter the human body by breaks, mucous membranes and punctures like the nose and mouth. Also people who prick or cut themselves by handling these waste are infected by the bacterial pathogens like *Corynebacterium diphtheria*, *Staphylococcus* spp., *Pseudomonas* spp. and *Escherichia coli* (Egbenyah et al., 2021).

MW has heavy metals in their disposal which consists of high density and is more toxic even when the quantity is less. Some heavy metals are lead, chromium, cadmium, mercury, thallium, etc. that affect the animal and human health when contaminated in land and water. The trace elements, namely zinc, selenium and copper, are also heavy metals that are needed for the human body to maintain metabolism; yet, they are toxic when they are obtained in higher concentrations. Mostly, the heavy metals in the human body are entered by air, drinking water and food (Emenike et al., 2022). There might be a negative impact on the improper disposal of MW on the quality of water as various pollutants leach out from dumping waste into the sites of MW (Ojha et al., 2022). Incineration of MW contains polycyclic aromatic hydrocarbons and heavy metals in high concentration which results in huge quantities of hazardous materials that pollute surface and health of individuals. Hence, there is a need for elimination of ash toxicity before disposing into reutilisation or landfills.

The ash of MW shows increased value in hardness and chloride content leachate that is above the permit limit set by Environment Protection Agency and World Health Organization (WHO) rules for drinking water (Okrikata & Nwosu, 2023). MW disposed of improperly and unscientifically changes the soil quality in the waste dumping areas. Various pollutants are mixed with the soil by dumping waste and lead to the change in the biology and chemistry of the soil ecosystem (Patnaik & Jena, 2021). Five heavy metals have been detected in the soil in the dumping areas of MW (chromium, nickel, zinc, lead and copper) (legalservice, 2023). As there is an increase in the population, the need for healthcare sectors has also increased, which leads to the MW. The MW that is disposed of improperly may affect the environment and health of the human in various aspects. Mainly water and soil are polluted by the MW that has an impact on the health of individuals.

While numerous reviews exist on medical waste management, this study is unique in its focus on the dose-response relationship of medical pollutants within soil and groundwater matrices. The scientific value of this review lies in its synthesis of global remediation technologies (2020–2025) and their applicability to the specific socio-economic and environmental constraints of developing nations like India.

Purpose of the study

The research aims to establish the dependency between waste disposal duration and the migration rate of pathogens in sub-surface environments. The study seeks to fill the knowledge gap regarding how "non-hazardous" waste (75–90% of MW) often serves as a primary vector for hazardous leachate when co-disposed. Our hypothesis is that current management policies fail because they overlook the cumulative exposure time of micro-pollutants; thus, this review aims to develop a pattern-based risk assessment model for future field research.

This review analyzes the environmental degradation of soil and water systems resulting from improper medical waste (MW) disposal. Moving beyond a general overview, the study specifically evaluates the role of hospitals as primary waste generators and examines the environmental impacts through the lens of pollutant concentration and exposure duration. The following sections synthesize current literature on waste generation ("Medical waste generation" section), the specific contamination pathways in water and soil ("Impact of medical waste on water" and "Medical waste impact on quality of the soil" sections) and hospital-specific management protocols ("Role of healthcare sector in managing medical waste" section). Furthermore, this paper identifies systemic failures in current mitigation strategies ("Mitigation options in controlling medical waste" and "Reason for failure of medical waste management" sections) and provides a critical analysis of the long-term ecological risks to flora, fauna and human health ("Critical analysis" and "Research gap" sections).

Medical waste generation

Medical waste (MW) is primarily generated in hospitals, dental clinics, pharmacies, healthcare centres, laboratories and blood banks (Torkashvand et al., 2022). While smaller clinics and households contribute to the waste stream, hospitals remain the primary point of origin for high-volume hazardous waste, requiring specialized on-site management protocols to prevent environmental leakage (Ravindra et al., 2023). MW contains animal or human tissue, excretions, syringes, sharps, needles, body fluids, dressings and pharmaceutical products; contact with any of these is proven to be hazardous (Agamuthu & Barasarathi, 2021).

Classification of medical waste

The waste is segregated priority wise; first, it is segregated based on non-reusable and reusable, non-hazardous and hazardous material. There are generally 4 steps in managing MW (Chand et al., 2021).

Classifying MW into safe and reusable materials in the proper container

- Disposal sites and waste treatment transportation
- Treatment
- Final disposal

WHO states that nearly 75% to 90% of MW are non-hazardous while the remaining are hazardous. Further, the hazardous waste is classified based on the risk of infection transferred during the process of managing waste. The WHO-standardized classification categorizes hazardous waste into four primary groups: (1) infectious which has blood fluids, intravenous (IV) lines, blood and dressings; (2) sharps waste which includes needles and scalpel blades; (3) pathological waste which includes body parts, microbiological cultures and anatomical body parts; 4) other waste which includes polyvinyl chloride (PVC) plastic, radioactive waste and mercury (Selvan Christyraj et al., 2021). The non-hazardous waste is non-infectious waste like municipal solid waste (Attrah et al., 2022). Figure 1 displays the types of hazardous medical waste.

Threat in improper disposal

The MW poses a great threat in the life quality of the healthcare workers. The MW results in disease or injury of healthcare workers, if waste is not managed properly. The following characteristics

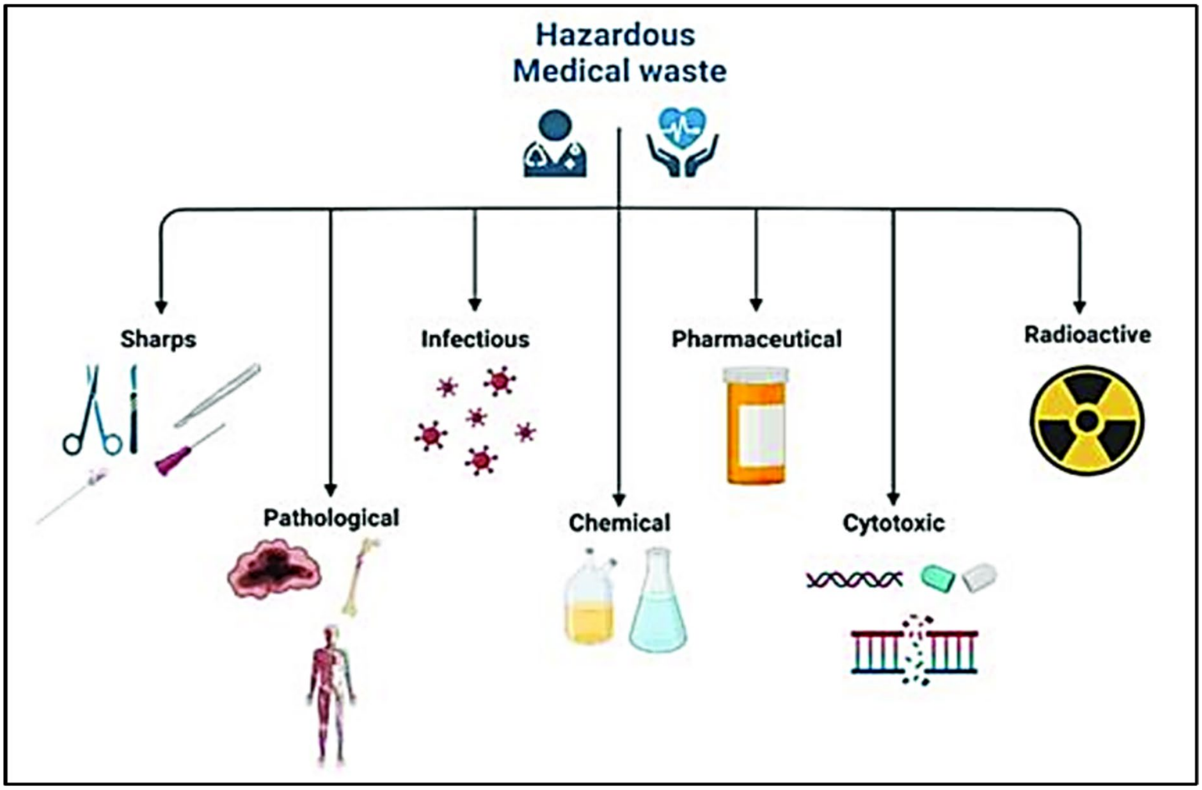

Fig. 1 Hazardous medical waste (Attrah et al., 2022)

are the nature of the healthcare waste; it contains genotoxic, sharps, infectious agents, radioactive and pharmaceuticals or hazardous chemicals (E. Janik-Karpinska et al., 2023a, 2023b). There are microorganisms that are harmful to the workers in healthcare sectors; other risks from the health facilities to the environment by improper disposal of MW is spread of drug-resistant microorganisms (Aleem et al., 2021). The risk in the health of humans related with the MW and their by-product are poisoning and pollution via pharmaceutical products, air pollution by the toxins released during incineration, sharps-inflicted injuries and many more (Ravindra et al., 2023). Dumping of medical waste contaminates the groundwater if not constructed properly. The incineration is practiced globally, but sometimes this process releases harmful pollutants into the air (Xu et al., 2023).

Improper disposal of MW has been a threat to domestic and wildlife animals, as the dumping of MW contaminates the water. The consequences of dumping MW are listed below.

- There is a threat in the spread of COVID infection among the community as thousands of individuals dispose of the face mask in the street.
- There is a high chance of negative threats to flora and fauna from the soil and GW pollution caused by the improper disposal of MW.
- As the water has been contaminated with Viagra there has been an uncontrolled sex drive among the sheep.
- Teenagers are lead to drug abuse by opioids disposed improperly; children are poisoned accidentally by the drugs disposed improperly, as they think pills are candy that are colourful also kids take medicine unintentionally that results in overdosing.
- There are many chances of getting the MW into the misusing hands in the public by selling contaminated drugs into a newly packaged drug.

- The MW has affected aquatic life by the change in the sexual characteristic by the pollutants with Ethinylestradiol. There has been a reduced hatching of eggs in the frogs and damage of organs in fish due to the contamination of Analgesics in water.
- There has been a decrease in the growth of the algae and aquatic plants as there is an increase in antifungal and antibiotic contents that are added to the water by improper disposal.

Methodology

This study adopts a comparative review methodology to analyze the impacts of medical waste disposal on soil and water. Rather than conducting primary field experiments, this research synthesizes data from existing case studies to evaluate the environmental outcomes of different waste management strategies.

Study design: comparative literature framework

To satisfy the requirement for impact analysis, this study categorizes existing research into two conceptual frameworks:

Improper disposal scenarios (case studies): Literature focusing on healthcare facilities or regions where waste is disposed of in unlined landfills, open dumps, or via illegal dumping.

Standardized management scenarios (reference framework): Literature focusing on facilities adhering to WHO and national standard operating procedures (SOPs), including proper segregation, autoclave treatment and engineered landfilling.

Parameters for impact analysis

The analysis of soil and water pollution is conducted by reviewing reported concentrations of specific pollutants across the selected literature. The study focuses on:

Physicochemical and heavy metal indicators: Assessment of lead (Pb), mercury (Hg) and cadmium (Cd) leaching into soil and groundwater.

Biological pathogens: Presence of enteric pathogens and multi-drug-resistant (MDR) bacteria in water bodies adjacent to disposal sites.

Exposure variables: To address the reviewer's concern regarding dose-response, this review synthesizes data regarding leachate concentration and the duration of environmental exposure reported in longitudinal environmental studies.

Data extraction and synthesis

Data was extracted from peer-reviewed journals (2018–2025) to ensure relevance. The synthesis focuses on:

Soil quality indicators: pH levels, heavy metal accumulation,and soil toxicity.

Water quality indicators: Chemical oxygen demand (COD), biological oxygen demand (BOD) and the presence of pharmaceutical residues (e.g. Ethinylestradiol, Analgesics).

Community health correlation: Summarizing observed health impacts in populations residing near high-contamination zones compared to those near facilities with "Standardized Management".

Impact of medical waste on water

By dumping the MW, the GW gets polluted and affects various diseases. Various aspects of the environment like wastewater, water quality of GW, soil and air quality need to be protected (Thakur et al., 2021). Disposal of medical waste in water sources may cause large pollution. This polluted water is mere unfit for agriculture and drinking purposes. This polluted water destroys the life of aquatic plants and animals. There is a seep in the polluted water via ground that pollutes the GW (Parida et al., 2022). The unplanned disposal of MW represents a high risk in the ecosystem of water, soil and air. The non-degradable substances contaminate both water and soil in the environment. In MW, mostly the microplastics like masks, gloves and other plastic supplies in the healthcare industry provide intensive pollution in water and soil (El-Ramady et al., 2021). Proper waste management is required in the healthcare centres. There is no adequate research related to the proper technique for disposing of the MW (Zhu & Wang, 2024). Moreover, MW have an effect for the long term on the health of human. Though the GW is polluted by the improper disposal, either by draining the waste or burying it in the ground.

The researchers have found that MW in domestic sewer system infects many individuals or animals by the contamination of water (Rahman & Ahmed, 2024). Mostly the liquid waste in the hospitals is drained. Many developing countries use uses wastewater for their agricultural purposes. There is no proper awareness in the developing countries regarding the toxic chemicals in the wastewater. The existing study discusses on the human health risk related to the water pollution (Moloantoa et al., 2022) (Arif et al., 2020; Shahid et al., 2020). Usually, the underwater contains heavy metals like calcium, sodium, free ions and dissolved salts; the rate of concentration of this metal increases with the disposal of toxic substances in the water. In most of the developing countries, the quality of the water does not meet the standard water quality of the country; this poses many health issues among millions of people in the country (Sultana et al., 2023). Global trends indicate that pharmaceutical substances entering the environment from medical facilities pose a growing threat to aquatic ecosystems, necessitating advanced monitoring of endocrine disruptors (Yerima & Atoshi, 2023). In the Indian context, the relationship between surface water quality and localized pollution sources is critical for ensuring public health (Muhammad et al., 2024).

Types of heavy metal identified in water

Various heavy metals pollute the underwater that is stated at the global scale (Zhang et al., 2023). There has been many research on the heavy metal concentration in the underwater at the chronic level, hence affecting people's health. Various anthropogenic activities and dumping of waste increase the contamination of the GW (Essien et al., 2022). The study of Kada estimates the heavy metal levels in the GW (Kada et al., 2022). The heavy metals like lead, copper, cadmium, nickel, manganese, chromium, iron and zinc are analyzed in the study. The findings of the study suggest that the due to the contamination of the water, many dangerous health risks have affected the individuals. The finding prevailing study also states that drinking heavy metal polluted water causes sub-chronic effects that decrease the oxidative stress, gastrointestinal ulcer, immunity level and also cancer (Kontoghiorghes, 2025). Numerous human diseases are mainly based on the exposure of the toxic metal specifically in blood and kidney, cardiovascular and nervous system disorders. The conventional study illustrates that waste contaminated water used for agriculture purpose eventually injects the toxic material in the water into the vegetable or food cultivated resulting in the decrease in the vital nutrients for the human body and initiating issues related to health that affect the organs in the human body (Abdullahi et al., 2021). The study further states that by consuming the foods cultivated by the contaminated water leads to infertility, failure in the reproduction, demineralization of bone, gastrointestinal cancer and impaired immunological defences.

Mostly the underwater and surface water are used for recreational and drinking purposes. In developing countries, lack of management is based on financial constraints, absence of environmental abatement measures like leachate polluted toxic liquid and water lining materials and collection systems (Puška et al., 2022). Improper measurement of leachate leads to toxins entering the GW that have a negative impact on the quality of water. This poses a risk to the ecosystem of plants and animals that use the polluted water for drinking purposes. It has been projected that cholera, typhoid, bacillary dysentery and amoebic infections are contributed by waterborne pathogens. Eighty percentage of the diarrhea in developing countries is due to the water polluted water that has improper disposal of MW (Nyekwere, 2021). The study of Nyekwere has shown that toxic liquid called leachate contaminates water and reduces the drinking water quality standards. The study of Eminike states that water pollution by the MW is based on radioactive, biological and chemical substances. The improper disposal of this waste leaches out by the dumping of waste into the GW (Emenike et al., 2022).

Literature review of MW disposal on water

Similarly. the study of Singh et al. (2024) states that there is an increase in the pollution and health-related issues in the MW. The waste in the hospitals is mainly due to sharps and the radioactive items. The throwing of items like bandages causes infections and spreads among the individuals. The study concludes that proper management is required for the disposal of medical waste. The study of Mailina discusses that the chemical waste, sharps waste and pharmaceutical waste are the major sources of waste in the healthcare activities and this waste must be disposed of properly by the method of incineration to avoid pollution (Mailina & Zakianis, 2021). Likewise, the study of Shiv Bolan has aimed at eliminating the waste in hospitals and analyzed the impact of the environment on MW disposal by poor management. The waste like plastic, adhesive plasters and battery waste are common types of waste (Bolan et al., 2023). The investigation of Gutu's research finds that the water pollution caused by the MW is due to the anatomical parts, as these parts have direct contact with the water during experimentation activities that further reach the sewer system (Guţu et al., 2022). This water gets polluted and also pollutes the water basins in the nearby localities. The study of Mustafa has found that a greater part of MW based on anatomical waste like teeth is produced by the dental clinics (Mustafa & Lahu, 2022). The research of Janik-Karpinska aims to present a view on MW stating that the HCW are potentially exposed to the risk regarding the improper disposal of MW (Edyta Janik-Karpinska et al., 2023a, 2023b). The study finds that there is an increase in waste in sharps and needles that are not disposed of properly. The study suggests that there is a need for awareness of the disposal techniques among the HCW to avoid impact on the environment due to MW.

Medical waste impact on quality of the soil

The HCW exhibit health issues in the developing countries regarding the MW. Among the MW, the sharp waste is the most dangerous waste. The study (Natasha et al., 2023) has highlighted that the contamination of chemical toxins in the wastewater has positive and negative impacts on the wastewater. The study suggests that accumulation of toxic materials in the soil has the possibility of transfer from plant to human; also, the study reveals that there are health hazards in the utilization of untreated waste for cultivation of vegetables or foods. Improper disposal of MW changes the quality of soil in the dumping sites of MW. Various pollutants get incorporated in the soil that changes the biology and chemistry of the soil. Many researchers have analyzed five important heavy metals in soil such as chromium, nickel, zinc, lead and copper. The soil in the dumping sites consists of higher concentrations of these metals. The Toxicity Characteristic Leaching Procedure (TCLP) is used to analyze the pollution potential in the soil in the dumping area and ashes by incinerator (Deepak et al., 2022).

Transmission of disease through MW

MW dumping is a good source of growth of microbes, multiplication and generation of rats and insects. Even though waste has proper moisture and nutrients, it is harmful and not suitable for a healthy environment. Many diseases have been spread worldwide due to the dumping of the MW such as trachoma, paratyphoid fever, cholera, typhoid, tuberculosis, bacillus and amoebic diarrhea, anthrax and plague. Waste dumping in the cities provides a place for maggot growth (Addy et al., 2023). Mostly, homes and other activities related to health dispose of their waste along with the domestic or household waste. Rodents such as rats may spread many diseases to humans. Rodents need food, shelter and water to reproduce that are in the waste (Dehghani et al., 2021). Few diseases are spread by rats that include plague and typhus fever. The disease leptospirosis affects both animals and humans. This occurs by eating contaminated food in the waste dumping areas. The rats play an indirect role in the disease transmission like amoebic diarrhea and *Trichinella spiralis* and Taenia transmission (Bhunia et al., 2022).

MW disposal discharges toxic chemicals into the soil environment and water. These toxic chemicals enter directly into the surface of the soil and contaminate. The occurrence of heavy metals like lead, arsenic, mercury and cadmium in the disposal of MW enters into the soil body and reduces the crops and plant growth after consuming the soil (Bano, 2025). The quality of soil is affected majorly in areas and nearby areas of waste disposal. The nutritional quality of maize crops is decreased and related health risk by consuming these contaminated crops has increased. The research has found that lysine and amino acid content is increased in the crops by 5.88 percentage (Liu et al., 2023). Additionally, plastic materials in the MW that cannot be decomposed easily make climate exchange in the environment of soil and soil organisms. There is a direct effect of water pollution based on the soil pollution in the areas that have effective emission of microbial and chemical pollution (Miyazono et al., 2025). Soil flexibility is an important element of quality and health of soil. After the pandemic, the degradation of soil has been a vital factor for insufficient human nutrition. As there has been a large amount of waste disposal worldwide, it is a need to implement circular economy in managing waste disposal (Lal, 2024).

Literature review of MW disposal on soil

The exposure of syringes leads to blood-borne pathogens; hence, the HCW should manage the MW effectively. The research of Gorcun aims on the assessment of used needle management for improving the safety system in the management of sharps in the MW (Görçün et al., 2023). The Gorcun study results that there is great disruption in the soil quality due to the insufficient management of MW. The investigation of Abdolinasab discusses the knowledge of dentists and practices in the management of dental waste in private and public dental offices and clinics. The study illustrates that a comprehensive management regarding the MW is necessary among the dentist in the developing countries (Abdolinasab et al., 2025). Similarly, the study of Ibadi discusses the risk of the MW disposal based on the poorly managed incinerators generating hazardous pollution to the environment. The research recommends that enhancements in the management and practices are required for a pollution-free environment (Ibadi et al., 2024). Similarly, the research results that by segregating the pharmaceutical wastes and the plastic waste from the health activity centre, the cost of transportation of waste disposal can be reduced. Also, the study finds that there is inadequate and uncontrolled discarding of incineration. This leads to health risks and environmental problems (Nduwimana et al., 2024). A major problem of MW is liquid waste in the healthcare sectors. There is no proper training and handling facility in the hospitals regarding the liquid waste. Effective management is required in handling the liquid waste to avoid the risk to the environment and humans (Majumder et al., 2021). The pandemic era mostly contains plastic materials like PPE kits, disposable masks, gloves and other associated protective wear that are not easily decomposed and disrupt soil quality. By the use of biological strategies and microbial effects, the removal of hazardous plastic waste during the period of the pandemic was practised in medical sectors (Dey et al., 2023).

In Fig. 2, the dumping of MW in the open areas is illustrated. The leftover chemicals in the bottles contaminate the land and water quality and help in spreading various diseases in surrounding areas.

Table 1 illustrates a comparative analysis based on the existing research types of medical disposal, the objective of the research and the results of the study.

Role of healthcare sector in managing medical waste

In healthcare sectors, the MW is separated in different colour containers based on the materials used. Table 2 explains the colour and the materials that are disposed and type of waste in the healthcare sectors.

The healthcare workers (HCW) should be involved in the proper disposal of MW. The HCW should be aware of the disposal methods in the required area. HCW should not allow disposal by burning, destruction and dumping disposal in the open sites as it may lead to risk to public health. The HCW should inform the authorities based on the availability regarding the disposal option, cost and the transportation of the disposal service (Mohammed et al., 2021). The HCW play a vital role in managing the MW to make sure of rational drug use to reduce the hazardous effect of MW. Proper policy, mechanism and rules are required for the disposal. Many researchers have revealed that pharmacists have better knowledge in waste disposal compared to other crews. There must be a track among the pharmacists to ensure the usage of the drugs to avoid waste (Mahamad Dom et al., 2023).

Mitigation options in controlling medical waste

The country's economy is mostly dependent on the extraction of natural sources and agriculture. The MW deploys these sources and contaminated

Fig. 2 Medical waste dumping in open areas (N. Singh et al., 2022a, 2022b)

them causing health-related issues among animals and humans. Providing reliable and safe management practice of medical waste is important to avoid issues regarding pollution and health-related problems. Proper decision-making process is required in segregating and disposing of the MW (Paengwangthong, 2022). Mostly, medical practices that are held in homes have MW that is combined with household waste. The awareness of disposing of MW should be created among every individual in the country. Additionally, during the COVID period, the clearance of used masks was along with the household waste (Sari et al., 2021). The guideline of WHO in disposing of the MW during the pandemic was segregating the non-infectious and infectious either in homes or healthcare centres and disposing of it properly (Hantoko et al., 2021).

Effective management of medicines and materials in the hospital is required among the working staff to avoid unnecessary MW (Leonard et al., 2022). Evaluation of knowledge on the disposal technique is required among the nurses (Dash et al., 2021). The policy suggestion regarding the MW is provided such as to enhance the discharge capacity and efficiency of MW, to contribute full play on advantages of policy based on hierarchical treatment and diagnosis and to advance the management level of MW in primary medical institutions (Hou et al., 2023). Recent advancements in 2024 and 2025 have identified several possibilities for the targeted removal of complex pollutants from hospital effluents. Techniques ranging from advanced oxidation to bio-filtration have shown efficacy in neutralizing diverse medical contaminants before they reach municipal systems (Belay et al., 2025; Yerima et al., 2024; Yue et al., 2020).

Table 3 provides various techniques regarding the disposal of MW.

Reason for failure of medical waste management

The reason for the failure of managing the medical waste is identified based on three barriers (Debrah et al., 2022).

- Lack of environmental education and protection
- Lack of market demand and pressure
- Lack of policy and regulatory process

The rate of production of wastage in health in the developing countries is more when compared with the developed countries (Tushar et al., 2023). Also, the developing countries suffer from a lack of specialists in the hospital and have untrained personnel, insufficient funding and lack of awareness regarding wastage (Banerjee et al., 2023). There are no practicing activities related to effective management of MW disposal; also there are no rules regulated in cooperating the waste management (Pattra et al., 2023). Additionally, the high capital cost for disposing of the MW is considered to be the key criterion for the reason of failure in the MW management (Reddy et al., 2023).

Table 1 Comparative analysis on the existing literatures

S.no	Reference	Type of medical waste	Type of environment affected	Objective	Conclusion
1	Singh et al. (2024)	Sharps and radioactive items	Water	The study outline effective strategies and policies for managing biomedical waste to enhance public health and promote a cleaner, safer environment	Implementing comprehensive biomedical MW strategies and policies is essential for reducing health risks and environmental pollution
2	Emenike et al. (2022)	Medical waste; focuses on heavy metal pollution in aquaculture	Water	To investigate the sources and impacts of heavy metal pollution in aquaculture and propose mitigation techniques	Effective mitigation techniques are essential for reducing heavy metal pollution and its negative effects on aquaculture systems
3	Nyekwere (2021)	Toxic liquid	Water	The study main goal is to minimize the hazardous waste	The author concludes that proper waste management strategy is required for disposing the hazardous waste
4	Mailina and Zakianis (2021)	Chemical waste, sharps waste and pharmaceutical waste	Water	The study discusses on the medical waste during pandemic era	The study concludes with the requirement of waste management to prevent pollution and also the incineration process must be further handled to reduce pollution
5	Bolan et al. (2023)	Plastic, adhesive plasters and battery waste	Water and soil	The study has viewed on the toxic elements after disposing MW	Effective segregation method is required for the waste discarding
6	Guţu et al. (2022)	Liquid waste and anatomical parts	Water	The study has aimed in monitoring the drinking water quality for the safety of beings and environment	The study concludes that proper disposal of liquid waste must be practiced for minimizing the negative effect to the environment and beings by the liquid waste
7	Mustafa and Lahu (2022)	Infectious waste, hazardous waste and general waste	Water	The objective of the study is to assess the impact of hospital waste on environmental pollution and its repercussions on human health	The study concludes that improper management of hospital waste significantly contributes to environmental degradation and poses serious health risks to the surrounding communities
8	Edyta Janik-Karpinska et al. (2023a, 2023b)	Sharps and needles	Water	The study view on the principles and the management of MW	The study results that there is inappropriate practice of waste management
9	Görçün et al. (2023)	Syringes	Soil	The objective of the study is to evaluate and select sustainable logistics that manage dispose of medical waste in an environmentally responsible manner	The study states the importance of enhancing waste management by reducing environmental impact

Table 1 (continued)

S.no	Reference	Type of medical waste	Type of environment affected	Objective	Conclusion
10	Abdolinasab et al. (2025)	Dental waste (anatomy waste)	Soil	The study examines the dentist knowledge on the waste management practice	The study results there is less awareness in the waste management practice among the dentist in developing countries
11	Ibadi et al. (2024)	Incinerators (anatomy waste)	Soil	The study assess the health risk of medical waste	The study suggest that proper practice in segregation and practice of medical waste is required among HCW
12	Nduwimana et al. (2024)	Sharps	Soil and water	Assess current practices of medical WM and identifying challenges and proposing improvements to enhance safety and compliance	The study concludes that inadequate medical waste management practices pose significant risks to health and the environment
13	Majumder et al. (2021)	Liquid waste	Soil and water	The study is based on the removal and persistence of the micro pollutant and infectious virus in the MW	The study concludes that development of technology would help to stop the pathogenic entities contaminating the environment
14	Dey et al. (2023)	Plastic waste	Soil	The study discusses on the impact on the micro plastic in the pandemic era	The study's outcome states that there has been lack in the practice and management of the waste, proper effort needed to be taken in a cost effective way

Table 2 Medical waste management in hospital (MWmanagement, 2021)

Colour of the container	Type of container	Type of waste	Treatment or disposal options
Yellow	Non-chlorinated plastic bags	Animal anatomy Human anatomy Expired medicines Soil waste Clinical, micro, biotechnological waste Chemical waste Liquid waste	Plasma Incineration Deep burial Pyrolysis
Red	Non-chlorinated plastic bags or container	Tubing, bottles, urine bags, gloves and syringes without needles	Microwaving Autoclaving Hydroclaving Recycling
White	Translucent leak, puncture, tamper proof container	Sharp waste and metals	Dry or auto heat Shredding Mutilation Encapsulation
Blue	Cardboard boxes with blue marking	Glassware	Autoclaving/disinfection Microwaving Hydrowaving Recycling

Critical analysis

Figure 3 illustrates the distribution of medical waste (MW) types and their proportional contribution to environmental loading. While anatomy waste (25%) and plastic/chemical waste (19%) represent the largest physical volumes, a critical impact analysis must consider the toxicity-exposure relationship rather than volume alone.

Chemical and pharmaceutical loading (dose-response)

Although toxic, chemical and pharmaceutical wastes constitute 19% of the total volume, they exhibit the highest environmental risk due to their persistence. The impact on soil and water is a factor of the leachate concentration (dose) and the permeability of the disposal site. Improperly, disposed pharmaceuticals (e.g. antibiotics and analgesics) create a continuous exposure loop in groundwater, leading to the development of drug-resistant microorganisms and endocrine disruption in aquatic life (Guţu et al., 2022).

Pathogenic and anatomy waste (exposure time)

Anatomy waste (25%) creates immediate biohazardous risks. However, the impact on soil quality is highly dependent on exposure time; as biological tissues decompose, they release high concentrations of nitrogen and phosphorus, leading to soil nutrient imbalance and the leaching of pathogens into shallow aquifers (Nduwimana et al., 2024).

Persistent pollutants (sharps and plastics)

Sharps (16%) and plastics (19%) pose long-term physical and chemical threats. In 2025, the degradation of pandemic-era PPE into microplastics has become a significant vector for heavy metal transport in soil. The cumulative exposure time of these non-biodegradable materials leads to the long-term leaching of additives (e.g. phthalates) into the groundwater table (Mailina & Zakianis, 2021).

Radioactive and liquid waste

While radioactive waste is the least frequent (9%), its "dose" impact is disproportionately high, requiring specialized decay-in-storage management. Liquid waste (12%), often discharged directly into municipal sewers, represents the most rapid pathway for contaminants to reach the aquatic "response" stage, affecting both human and animal health within short timeframes (Singh et al., 2024).

Table 3 Method of disposal of waste in the healthcare sectors (Kanyal et al., 2021)

Serial no	Waste content	Component	Method of disposal
1	Animal waste	Animal tissues, fluid organs, body parts carcasses, blood and animals used in research, bleeding parts, waste produced by veterinary hospitals, discharge from hospitals, animal houses	Incineration or deep burial
2	Human anatomical waste	Human organs, tissues, body parts	Incineration or deep burial
3	Discarded medicines and cytotoxicdrugs	Wastes related to outdated, contaminated and discarded medicines	Incineration, demolition and landfills
4	Microbiology and biotechnology waste	Wastes generated from laboratory cultures, specimens or stocks of microorganisms live or attenuated vaccines, animal and human cell culture used in research and infectious agents from research and industrial laboratories, toxins, wastes from production of biologicals, dishes and devices used for cultures transfer	Local autoclaving or microwaving or incineration
5	Solid waste	Items contaminated with blood, and body fluids including cotton, dressings, soiled plaster casts, lines, beddings, other material contaminated with blood	Incineration or autoclaving or microwaving
6	Waste sharps	Needles, syringes, scalpels blades, glass, etc. that may cause puncture and treatment cuts. This includes both used and unused microwaving and sharps	Disinfection (chemical treatment or auto claving or microwaving and mutilation or shredding)
7	Chemical waste	Chemicals used in production of biologicals, chemicals used in disinfection, as insecticides, etc	Chemical treatment or discharge into drains for liquid or secured landfill for solids
8	Liquid waste	Waste generated from laboratory and washing, cleaning, house-keeping disinfecting activities	Disinfection by chemical and treatment and discharge into drains
9	Solid waste	Wastes generated from disposable items other than the waste sharps such as tubings, catheters and intravenous sets	Disinfection by chemical treatment or autoclaving or microwaving and mutilation shredding
10	Liquid waste	Waste generated from laboratory and washing, cleaning, house-keeping and disinfecting activities	Disinfection by chemical treatment and discharge into drains

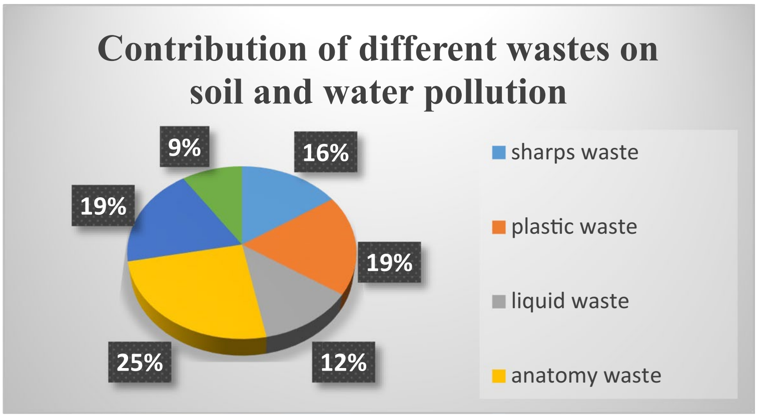

Fig. 3 Contribution of different waste on soil and water

Research gap

The study by Khalid et al. (2021) emphasizes administrative challenges like healthcare worker training; this review identifies a more critical scientific gap: the lack of longitudinal data quantifying the dose-response relationship between medical leachate and soil-water toxicity. Current research often focuses on waste volume rather than the long-term exposure time of specific pollutants like heavy metals and multi-drug-resistant pathogens in groundwater. There is a profound need for standardized environmental monitoring frameworks that track the migration of pharmaceutical active ingredients (PAIs) from improper disposal sites into the food chain. Future research must move beyond management techniques and focus on the bio-remediation of soil already contaminated by systemic medical waste mismanagement.

Emenike et al. (2022) identify heavy metals in MW as a significant health risk contributing to diseases. The current findings delve into the chronic effects of drinking water contaminated with heavy metals and discuss its implications for food safety and human health, which have not been well-explored in the existing literature.

The research by Chowdhury et al. (2022) mentions the link between improper MW disposal and various diseases but lacks a detailed exploration of specific case studies illustrating these connections. The present study provides real-world examples of diseases linked to waste mishandling, underscoring the urgency of addressing MW disposal issues comprehensively.

According to Shammi et al. (2021), a significant proportion of medical waste generated at the household level is often mixed with domestic waste. The findings of the current study address this gap by asserting the necessity for public education and community engagement initiatives to raise awareness about the need for proper segregation at the household level.

Several studies, including Nyekwere (2021), approach water pollution from a singular perspective without adequately discussing the interrelationship between soil and water contamination from medical waste. The current research fills this gap by systematically analyzing how various pollutants from medical waste affect both soil and water quality, thereby providing a holistic view critical for understanding environmental health dynamics.

Discussion

Your findings highlight several critical areas concerning medical waste (MW) management, particularly the need for enhanced training and awareness among healthcare workers. This aligns with the findings of Khalid et al. (2021) which emphasize the inadequate training that healthcare professionals receive, leading to improper disposal practices that pose health risks. By proposing comprehensive training programs, your research addresses this gap and aligns with existing calls for more structured educational initiatives within healthcare settings. Additionally, your focus on the long-term effects of heavy metal pollution, supported by Emenike et al. (2022), underscores the importance of understanding the chronic health implications of contaminated drinking water, an aspect that remains insufficiently explored in current studies. Both your work and Emenike et al. highlight the criticality of addressing heavy metal waste, thereby stressing the necessity for awareness among healthcare providers regarding waste handling practices.

Moreover, your findings introduce a significant emphasis on public education and community engagement to improve household medical waste disposal, an area that (Shammi et al., 2021) touch upon but do not fully develop. While they acknowledge the risks associated with mixing medical waste with domestic waste, your research advances the discourse by advocating for practical strategies that promote proper segregation practices at the household level. Additionally, the interrelationship between soil and water pollution from medical waste, as discussed in your research, complements Nyekwere (2021) perspective but goes further by systematically analyzing pollutants' effects on both environmental components. This holistic view enriches the discourse by providing a comprehensive understanding of the environmental health dynamics involved in MW management, ultimately advocating for integrated strategies that align with current sustainability initiatives in the healthcare sector.

Conclusion and future scope of research

This study systematically reviewed the pathways through which improper medical waste (MW) disposal facilitates the degradation of soil and water systems. Our analysis transitions from a conceptual understanding of waste management to a critical evaluation of environmental risk. The findings indicate that the impact of MW is not merely a function of volume, but a complex result of pollutant dose-response and cumulative exposure time.

Data synthesized from 2021 to 2025 reveals that pharmaceutical residues and heavy metals act as persistent environmental stressors; their "dose" in groundwater often exceeds safety thresholds near unlined disposal sites, leading to measurable biological "responses" such as endocrine disruption in aquatic fauna and the proliferation of antibiotic-resistant genes in soil microbiota. While current policies emphasize segregation, this review highlights a critical need for longitudinal exposure modelling to quantify the long-term migration of pathogens from burial sites into deep aquifers.

Furthermore, the transition toward a circular economy (CE) in the healthcare industry (HCI) must move beyond administrative "awareness" and focus on technical recovery. By 2025, innovative closed-loop systems such as the validated return-back channels for expired pharmaceuticals (Abbas et al., 2025) and high-efficiency manufacturing waste reduction (Dennison et al., 2024) are essential to mitigate the chemical loading of the environment. Future research should prioritize field-based quantification of heavy metal leaching rates over time to replace "idea-based" management with data-driven environmental protection strategies.

Author contributions All Authors (Rashi Yadav) 1) made substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data; or the creation of new software used in the work; 2) drafted the work or revised it critically for important intellectual content; 3) approved the version to be published; and 4) agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Declarations

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